Belief propagation and MPS for LR-QAOA

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Simulating and Sampling from Quantum Circuits with 2D Tensor Networks

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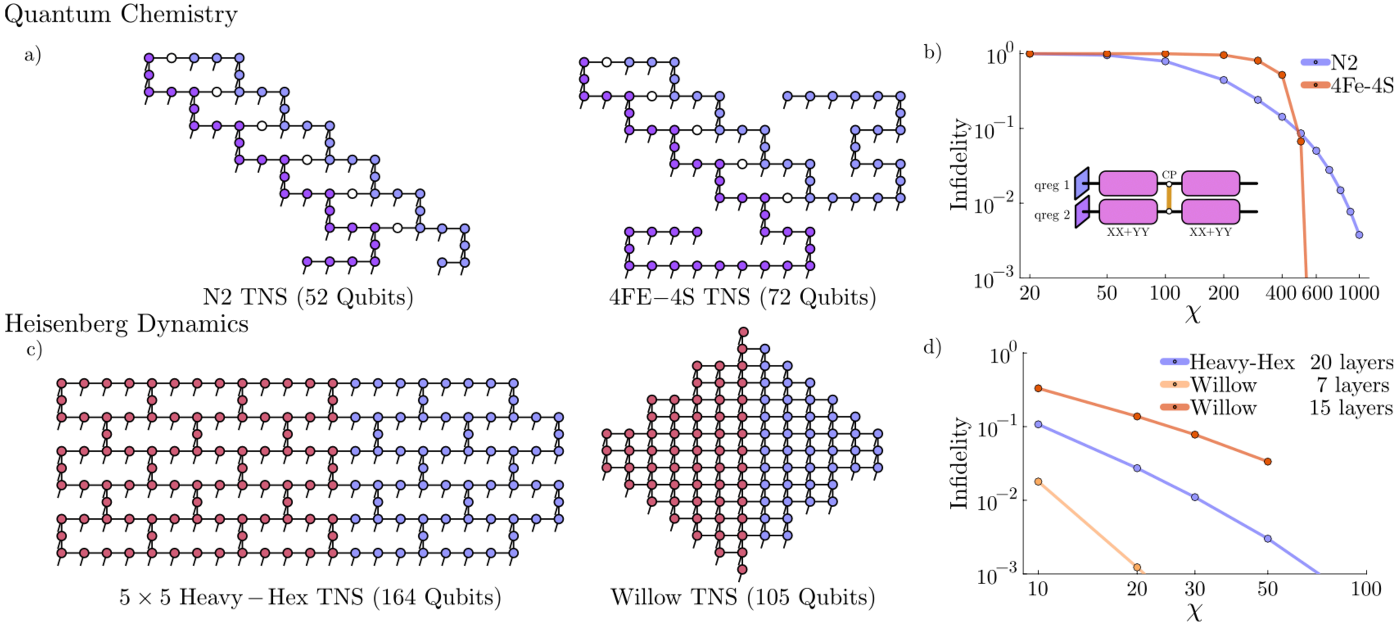
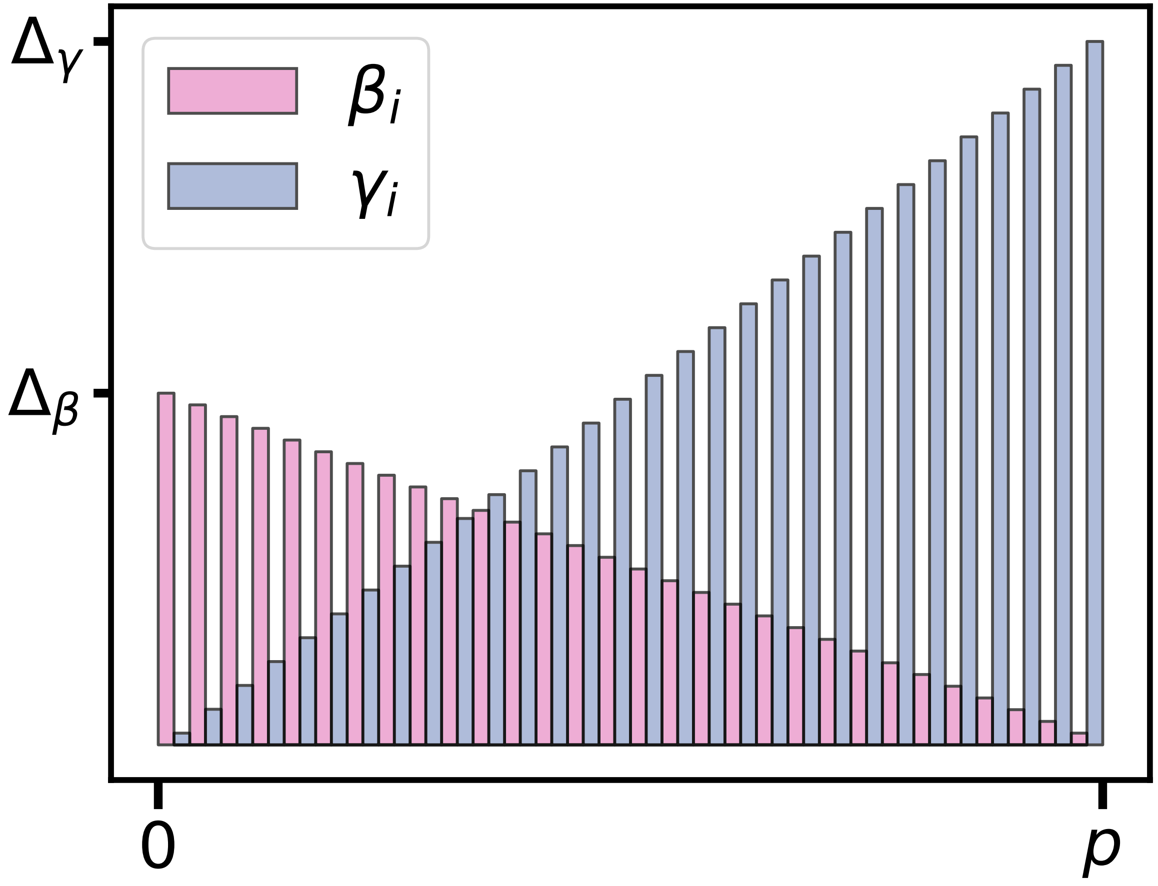
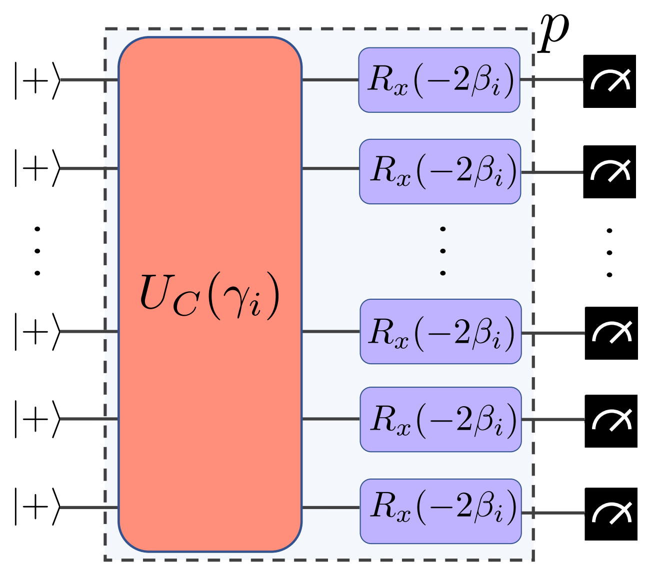


Figure 1. **Our tensor network topologies and their simulation errors.** a) The tensor network states (TNS) for simulating the N2 and 4FE-4S LUCJ circuits according to Ref. [12]. Vertex colors indicate quantum sub-registers which encode the different spin states for the considered electronic system. b) Approximate infidelity (c.f. $1 - f$ following Eq. (2)) of the final TNS as a function of the maximum bond dimension χ in our simulations. We also sketch the circuit structures, which consist of single-site gates and XX+YY rotations within the quantum registers, and CP gates between them. c) The TNS used for simulating discrete-time dynamics under the Heisenberg Hamiltonian on the heavy-hex or Willow processor topologies. Vertex colors indicate the regions of the model initialized in the 0- or the 1-state. d) Approximate infidelity of the final TNS after simulating 20 layers on the heavy-hex topology, and 7 or 15 on the Willow topology. Further details of the simulations can be found in Sec. III.

1. LR-QAOA

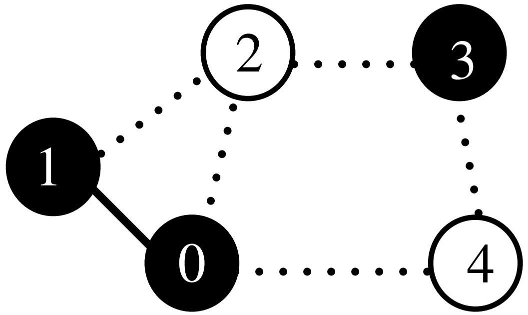


$$\beta_i = \left(1 - \frac{i}{p}\right)\Delta_\beta \quad \text{and} \quad \gamma_i = \frac{i+1}{p}\Delta_\gamma$$

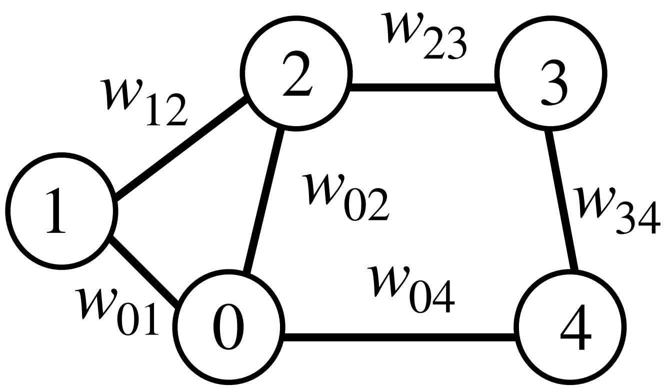


The problem behind LR-QAOA

$$x^* = 00101$$



The weighted maxcut (WMC) problem involves determining the partition of the vertices in an undirected graph so that the total weight of the edges between the two sets is maximized.



Cost function

$$C(x) = \sum_{(i,j)\in E} w_{ij}(x_i + x_j - 2x_i x_j)$$

Approximation ratio $\quad r = \dfrac{\sum_{k=1}^{n} C(x^k)/n}{C(x^*)}$

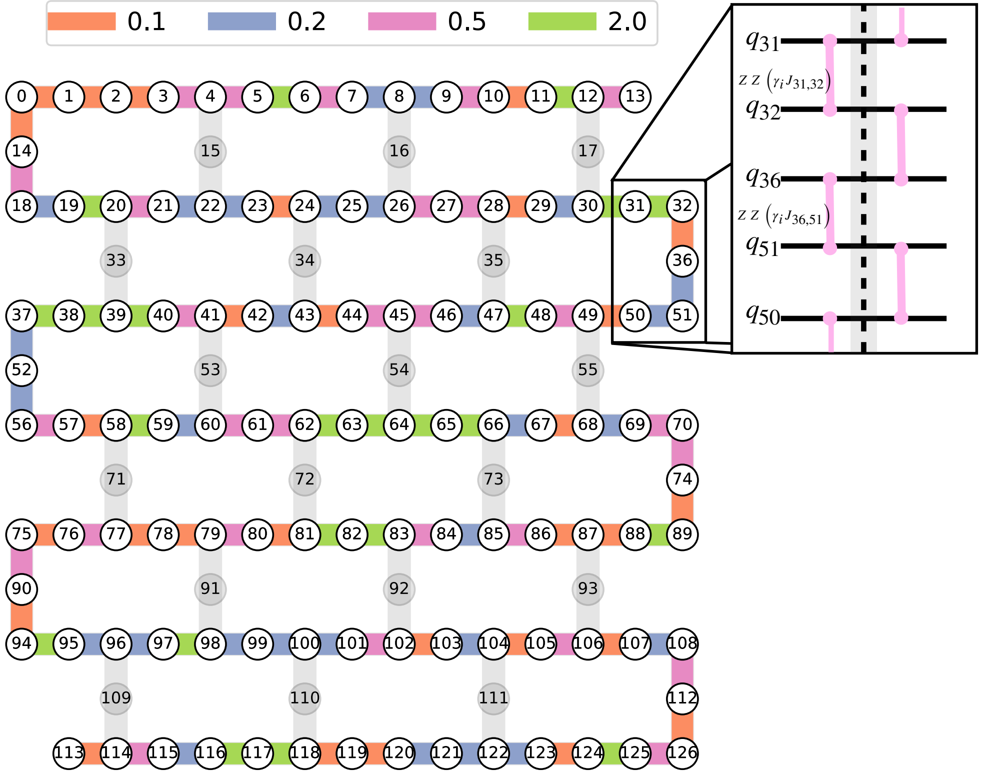
x_k sample solution

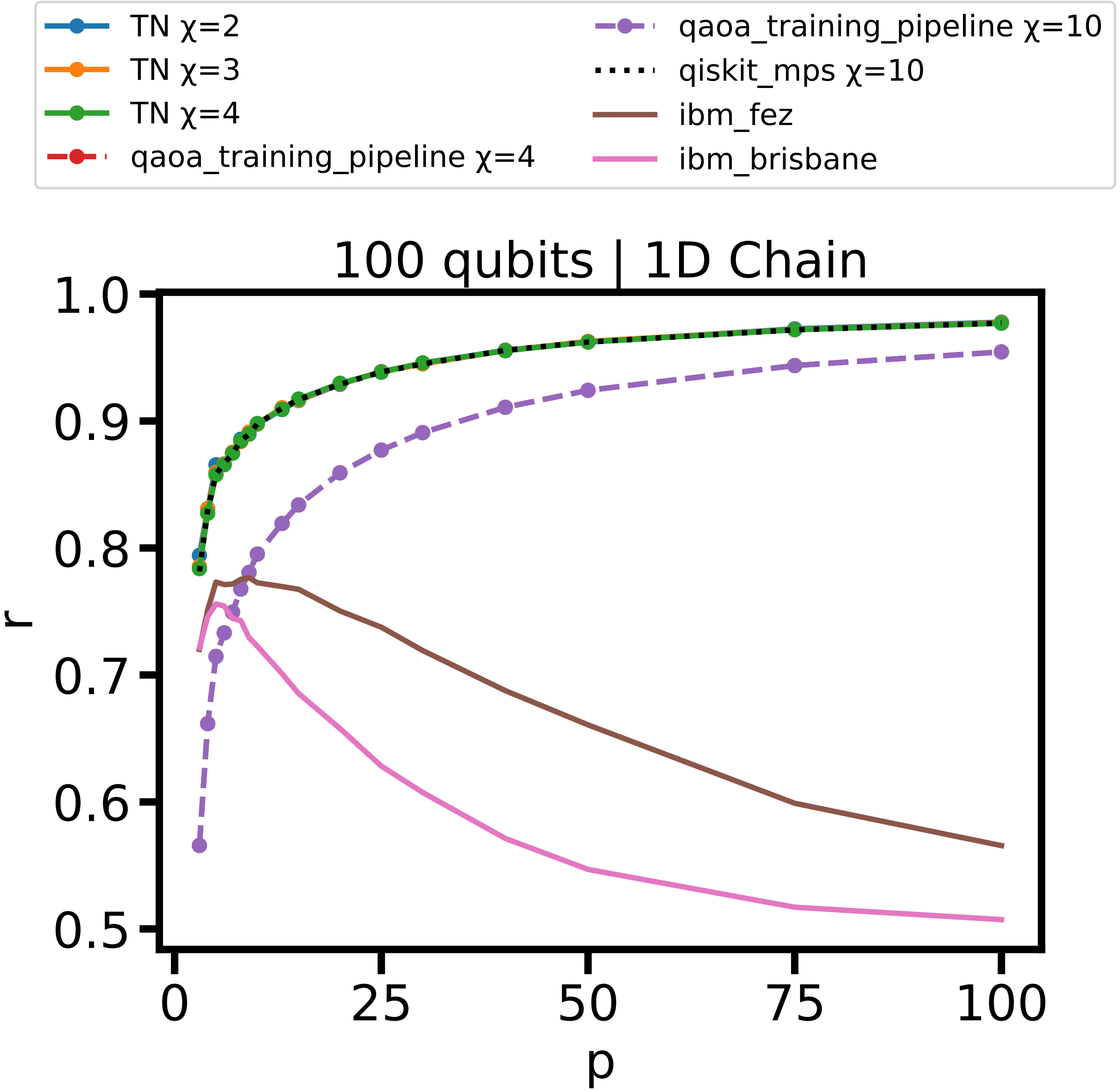
x^* optimal solution

n Samples

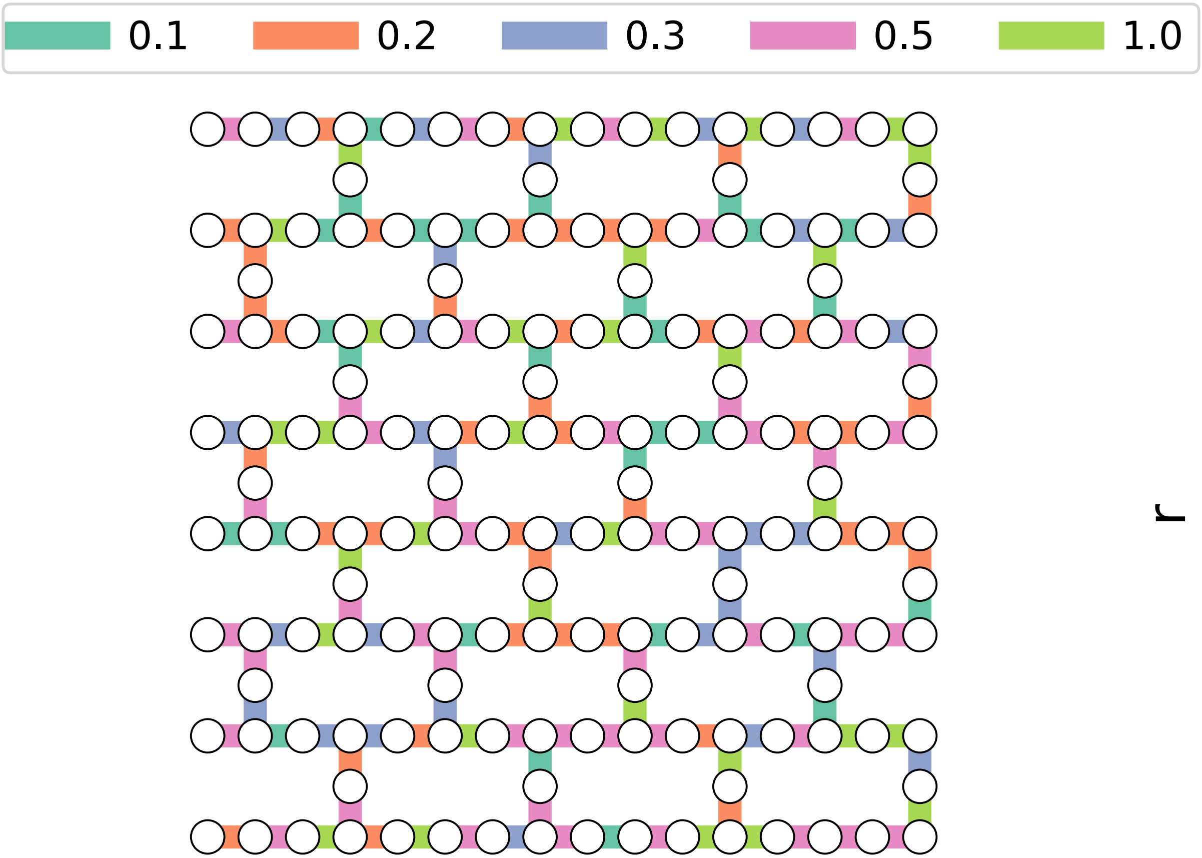
Performance metric

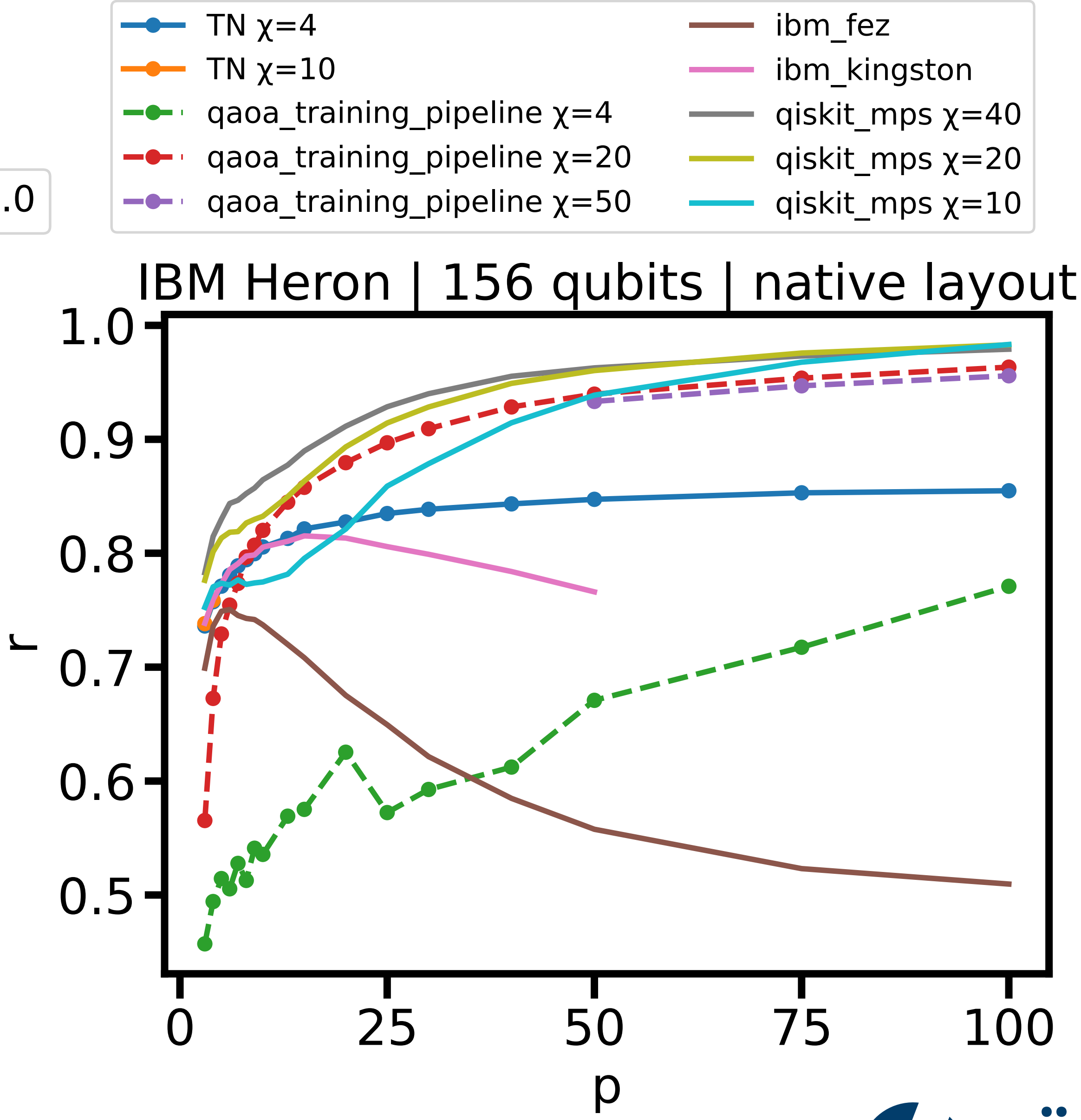
1D-Chain problem



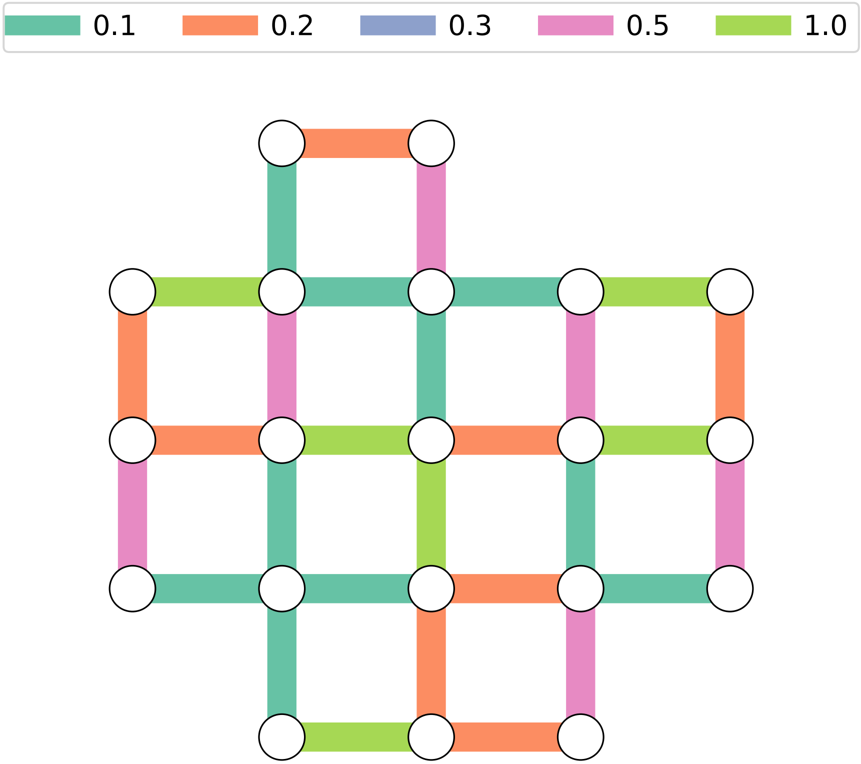


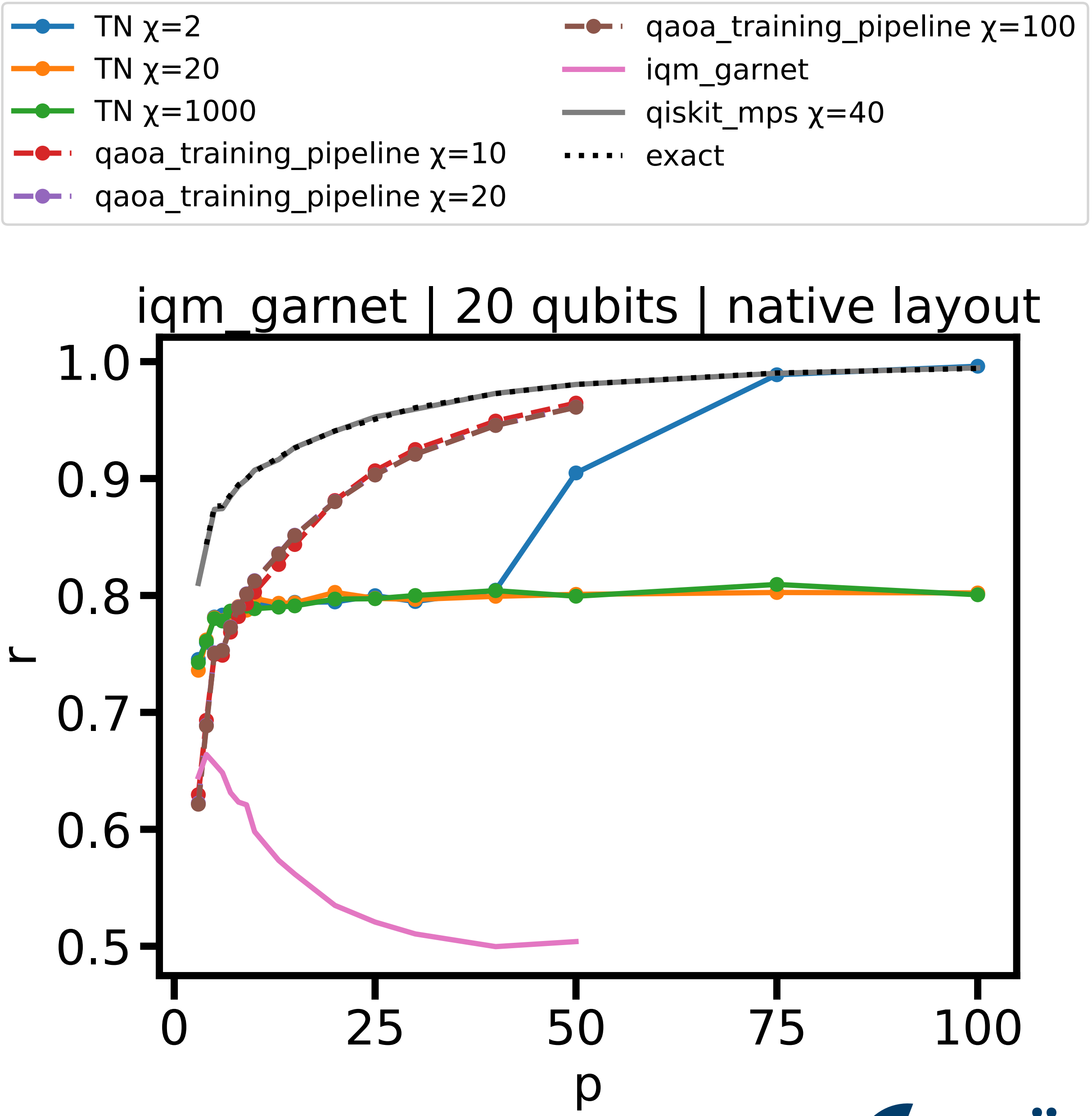
Heavy-Hex problem





Square lattice problem





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1D-Chain landscape - 30 qubit problem